

From: Lowe, Claire (EGLE)
Sent: 11/21/2024 4:33:51 PM
To: "Elaine J. Venema" <evenema@fveng.com>
Subject: RE: Cadillac MAHL revisions

Hi Elaine,

I discussed this question with the IPP committee. The consensus is that minimum removal shouldn't be used, and that if you want to be more conservative with removal efficiency then the decile method of calculating removal allows this. We would recommend complying with the EPA guidance on MAHL calculations, including use of actual plant flow and the most conservative NPDES limits.

Thank you for your patience on this response.

Claire Lowe

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MICHIGAN DEPARTMENT OF
ENVIRONMENT, GREAT LAKES, AND ENERGY



From: Elaine J. Venema <evenema@fveng.com>
Sent: Thursday, October 3, 2024 11:39 AM
To: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Subject: RE: Cadillac MAHL revisions

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Hi Claire –

We're working through the comments. Like you said, most are just easy fixes; but I do have a question on the NPDES based AHL calcs for the compatible pollutants. Back when Deb Quinn was at DEQ/EGLE in the IPP section (around 2010/2011 maybe), she required review of ALL of the NPDES permit conditions, not just the monthly average condition.

She had developed her own compatibles spreadsheet, which I based my standard MAHL template on.

Here's a screenshot of Deb's spreadsheet from a 2011 project for another client.

Blue cells contain entered data. Tan cells show results of formulas

Season:	Spring - Summer						
	Removal Efficiency, Min	Removal Efficiency, Avg	NPDES Limits Mo Ave	NPDES Limits Daily/7day	WWTP Flow MGD	MAHL, NPDES Mo Ave	MAHL, NPDES Daily/7day
	Decimal	Decimal	mg/l	mg/l		#/day	#/day
	(R _{potwm})	(R _{potwa})	(C _{npdes})	(C _{npdes7})	(Q _{potw})	(AHL _{npdes})	(AHL _{npdes7})
BOD-5	0.962	0.987	11	17	1.75	12350	6529
TSS	0.947	0.98	30	45	1.75	21893	12392
AMMONIA	0.753	0.965	9.5		1.75	3962	
P	0.815	0.898	1		1.75	143	
Season:	Fall - Winter						
	Removal Efficiency, Min	Removal Efficiency, Avg	NPDES Limits Mo Ave	NPDES Limits Daily/7day	WWTP Flow MGD	MAHL, NPDES Mo Ave	MAHL, NPDES Daily/7day
	Decimal	Decimal	mg/l	mg/l		#/day	#/day
	(R _{potwm})	(R _{potwa})	(C _{npdes})	(C _{npdes7})	(Q _{potw})	(AHL _{npdes})	(AHL _{npdes7})
BOD-5	0.94	0.987	25	40	1.75	28067	9730
TSS	0.931	0.978	30	45	1.75	19902	9518
AMMONIA	0.648	0.973			1.75		
P	0.777	0.912	1		1.75	166	

I think the thought process was that the WWTP must be able to meet the less restrictive concentration (like 7-day avg or max day) assuming a high flow date (Q_{max}) and lower removal efficiency. Then you just take the most restrictive AHL to ensure all conditions are covered.

If that is no longer the preferred way EGLE/Michigan would like to do this, please let me know and I'll focus on just the Monthly average concentrations with R_{avg} compared to the design basis.

Thanks,

Elaine

From: Lowe, Claire (EGLE) <LoweC9@michigan.gov>
Sent: Tuesday, September 24, 2024 2:26 PM
To: Elaine J. Venema <evenema@fveng.com>
Cc: utilities@cadillac-MI.net; Patrick Miller <pmiller@cadillac-mi.net>
Subject: Cadillac MAHL revisions

Elaine,

Sorry for the delay in getting comments back to you on the Cadillac MAHL. I do think we're close now. I have a few comments, attached, but I think mostly they are corrections on plugging the correct number into the calculation that shouldn't take long. Please let me know if you need to discuss anything on the attached further.

Thank you,

Claire Lowe

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